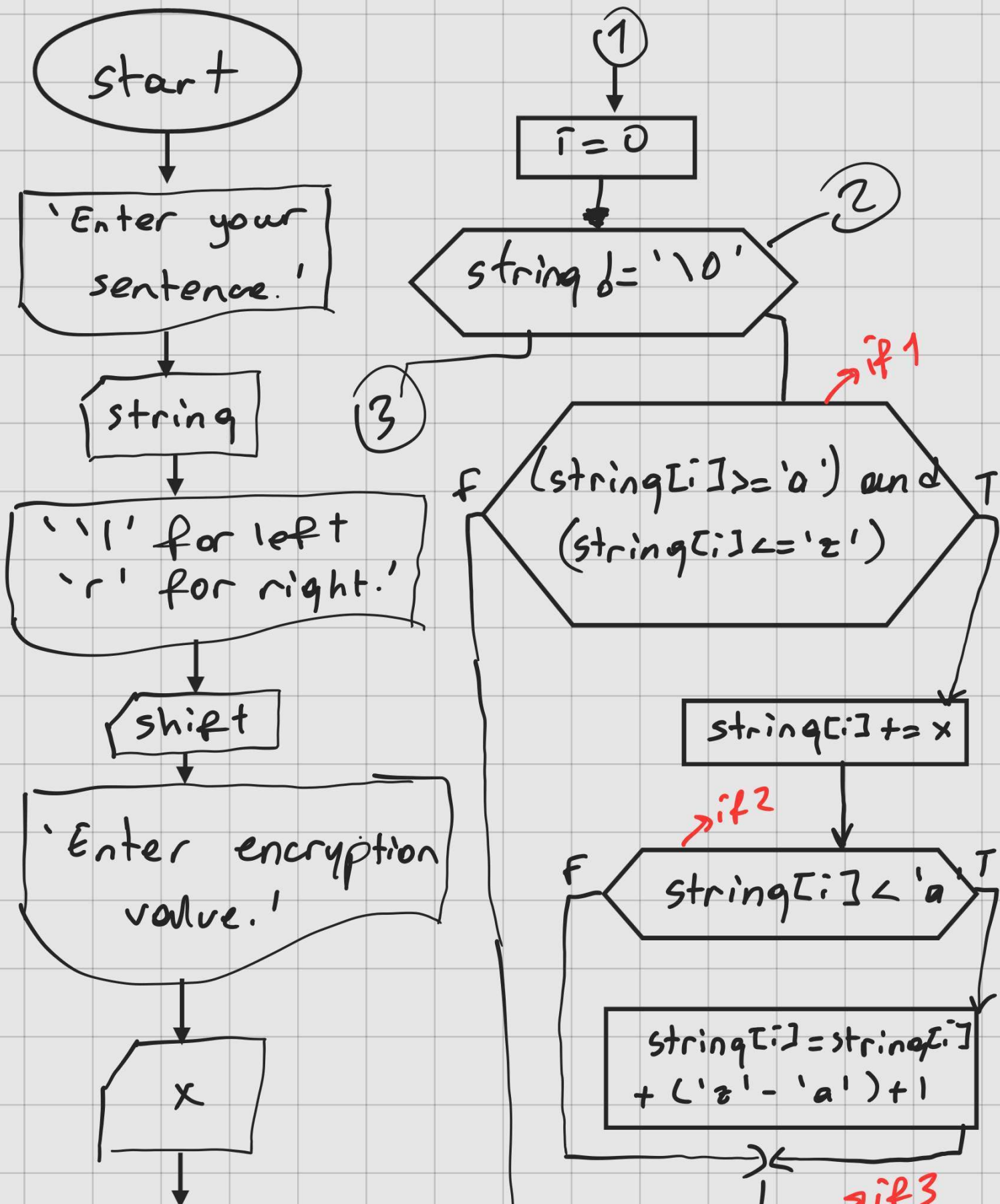


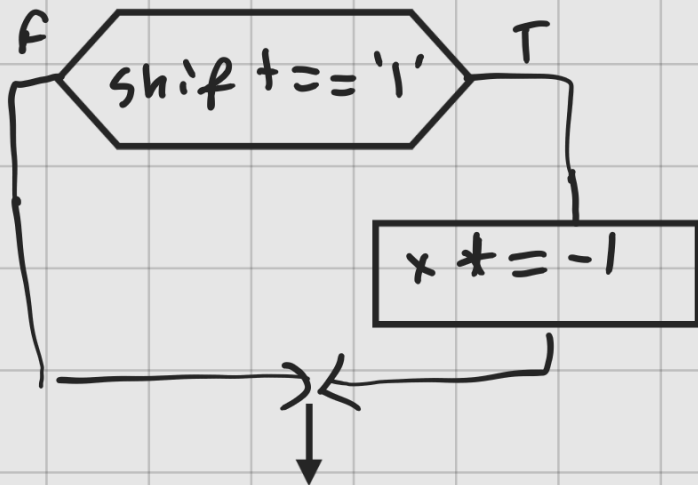
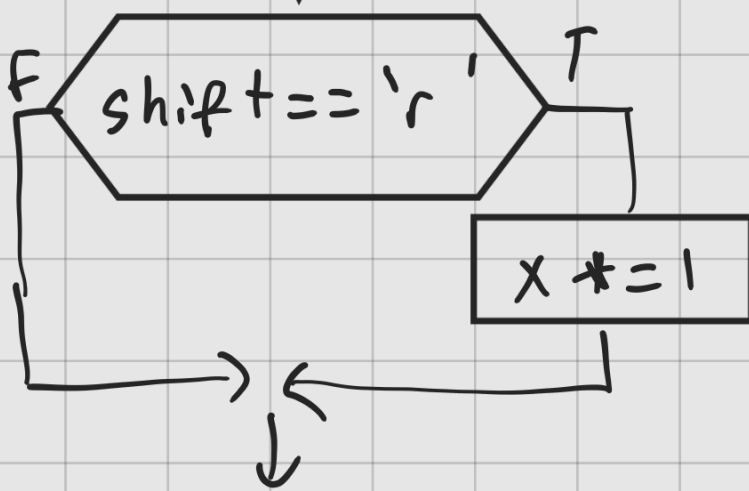
Flowchart + Analysis + Source Code.

Algorithm 1: Design an algorithm and write the program which encrypts the given text based on the following formula and decrypts it into the original form.

Shift = "left" -----> letter - X -> newletter

shift = "right" -----> letter + X -> newletter





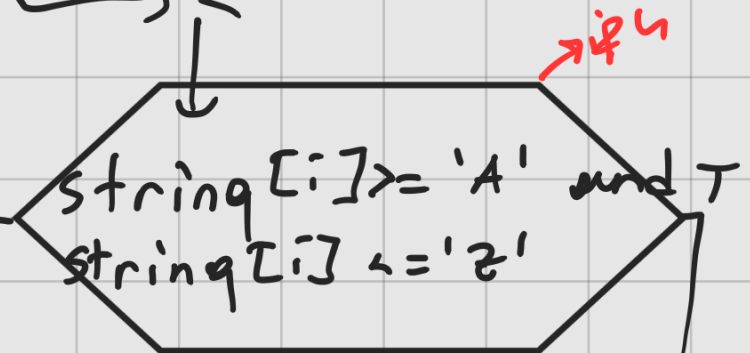
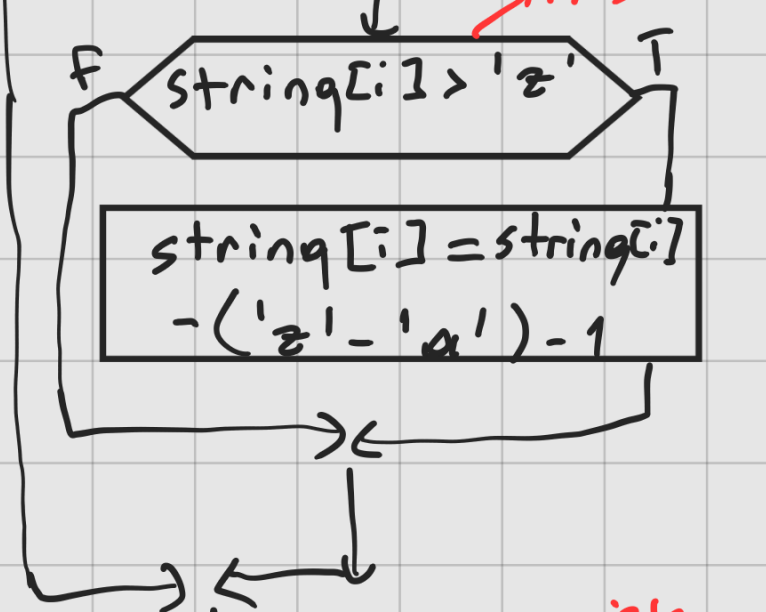
'Encryption is starting.'

(1)

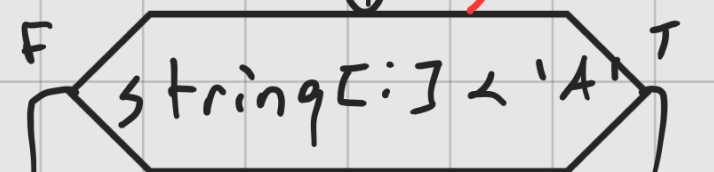
(3)

'Encrypted text is: ', string

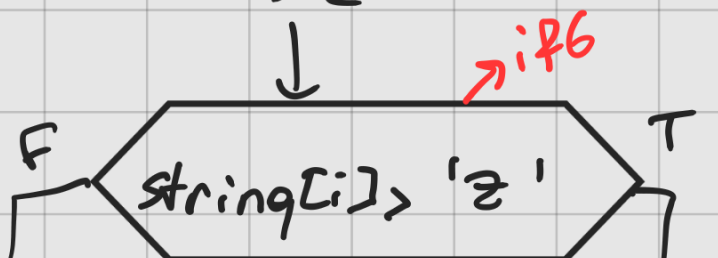
i = 0



string[i] += x



string[i] = string[i] + ('z' - 'A') + 1



string[i] = string[i]

$x = -1$

$-(\text{'z'} - \text{'a'}) - 1$

④
 $\text{string}[i] \neq \text{'\0'}$

$i++$

⑤
F $\text{string}[i] \geq \text{'a'}$ and $\text{string}[i] \leq \text{'z'}$ T

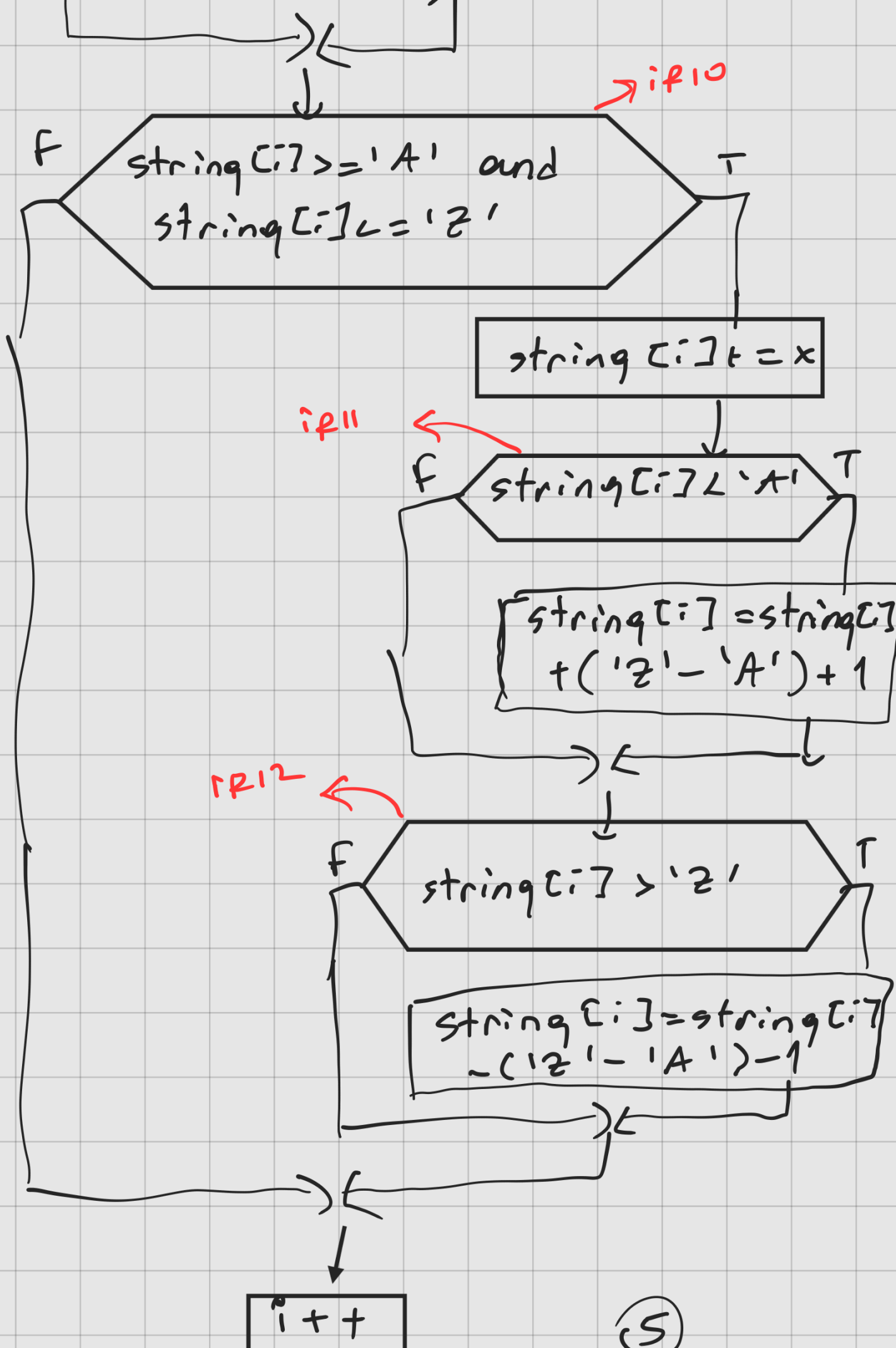
$\text{string}[i] += x$

F $\text{string}[i] < \text{'a'}$ T

$\text{string}[i] = \text{string}[i] + (\text{'z'} - \text{'a'}) + 1$

$\text{string}[i] > \text{'z'}$ T

$\text{string}[i] = \text{string}[i] - (\text{'z'} - \text{'a'}) - 1$



h

Decrypted text
is 'string'

Stop

Analiz

Example 1:

Text= I love this game.

Shift = left

X=1

Answer:

Encrypted Text: H knud sghr fzld.

Decrypted Text: I love this game.

i	shift	x	string[i]	Hangi ifte	işlem
0	1	-1	I → H	if 4	'I' - 1 = H
1	1	-1	-	x	x
2	1	-1	l → k	if 1	'l' - 1 = k
3	1	-1	o → n	if 1	'o' - 1 = n

4	1	-1	$v \rightarrow u$	if 1	'v' - 1 = u
5	1	-1	$e \rightarrow d$	if 1	'e' - 1 = d
6	1	-1	-	x	x
7	1	-1	$t \rightarrow s$	if 1	't' - 1 = s
8	1	-1	$h \rightarrow g$	if 1	'h' - 1 = g
9	1	-1	$i \rightarrow h$	if 1	'i' - 1 = h
10	1	-1	$s \rightarrow r$	if 1	's' - 1 = r
11	1	-1	-	x	x
12	1	-1	$g \rightarrow f$	if 1	'g' - 1 = f
13	1	-1	$a \rightarrow z$	if 1 + if 2	'a' - 1 = 'z' + ('z' - 'a') + 1 = z
14	1	-1	$m \rightarrow l$	if 1	'm' - 1 = l
15	1	-1	$e \rightarrow d$	if 1	'e' - 1 = d
16	1	-1	.	x	x
17	1	-1	null $\rightarrow$ while'down a.k.		

Encrypted text: H knud sghr fzd.

Decryption ↓

0	1	x*-1=1	$H \rightarrow I$	if 10	'H' + 1 = 'I'
1	1	1	-	x	x
2	1	1	$k \rightarrow l$	if 7	'k' + 1 = 'l'
3	1	1	$n \rightarrow o$	if 7	'n' + 1 = 'o'
4	1	1	$u \rightarrow v$	if 7	'u' + 1 = 'v'

5	1	1	d → e	if 7	'd' + 1 = 'e'
6	1	1	-	x	x
7	1	1	s → t	if 7	's' + 1 = 't'
8	1	1	q → h	if 7	'q' + 1 = 'h'
9	1	1	h → i	if 7	'h' + 1 = 'i'
10	1	1	r → s	if 7	'r' + 1 = 's'
11	1	1	-	x	x
12	1	1	f → g	if 7	'f' + 1 = 'g'
13	1	1	z → a	if 7 + if 9	'z' + 1 - ('z' - 'a') - 1
14	1	1	l → m	if 7	'l' + 1 = 'm'
15	1	1	d → e	if 7	'd' + 1 = 'e'
16	1	1	•	x	x
17	1	1	'\0' → while 'don't		

Decrypted text is: I love this game.

Example 2:

Text= How old are you?

Shift = right

X=2

Answer:

Encrypted Text: Jqy qnf ctg aqw?

Decrypted Text: How old are you?

<u>i</u>	<u>shift</u>	<u>x</u>	<u>string[i]</u>	<u>hang: if 'te</u>	<u>islem</u>
0	r	2	H → J	if 4	'H' + 2 = 'J'
1	r	2	O → Q	if 1	'O' + 2 = 'Q'
2	r	2	w → y	if 1	'w' + 2 = 'y'
3	r	2	-	x	x
4	r	2	O → Q	if 1	'O' + 2 = 'Q'
5	r	2	l → n	if 1	'l' + 2 = 'n'
6	r	2	d → f	if 1	'd' + 2 = 'f'
7	r	2	-	x	x
8	r	2	a → c	if 1	'a' + 2 = 'c'
9	r	2	r → t	if 1	'r' + 2 = 't'
10	r	2	e → g	if 1	'e' + 2 = 'g'
11	r	2	-	(x)	(x)
12	r	2	y → a	if 1 + if 3	'y' + 2 = (('z' - 'a') - 1) = 'a'
13	r	2	O → Q	if 1	'O' + 2 = 'Q'
14	r	2	v → w	if 1	'v' + 2 = 'w'
15	r	2	?	(x)	(x)
16	r	2	null	→ while 'dan' cik.	

Encrypted text: Jqy qnf cto aqw?
Decryption 1

0	r	$x - 1 = -2$	$J \rightarrow H$	if 10	'J' - 2 = 'H'
1	r	-2	$q \rightarrow o$	if 7	'q' - 2 = 'o'
2	r	-2	$y \rightarrow w$	if 7	'y' - 2 = 'w'
3	r	-2	-	(X)	(X)
4	r	-2	$q \rightarrow o$	if 7	'q' - 2 = 'o'
5	r	-2	$n \rightarrow l$	if 7	'n' - 2 = 'l'
6	r	-2	$f \rightarrow d$	if 7	'f' - 2 = 'd'
7	r	-2	-	(X)	(X)
8	r	-2	$c \rightarrow a$	if 7	'c' - 2 = 'a'
9	r	-2	$t \rightarrow r$	if 7	't' - 2 = 'r'
10	r	-2	$g \rightarrow e$	if 7	'g' - 2 = 'e'
11	r	-2	-	(X)	(X)
12	r	-2	$a \rightarrow y$	if 7 + if 8	'a' - 2 + ('z' - 'a') + 1 = 'y'
13	r	-2	$q \rightarrow o$	if 7	'q' - 2 = 'o'
14	r	-2	$w \rightarrow u$	if 7	'w' - 2 = 'u'
15	r	-2	?	(X)	(X)
16	r	-2	'\0' $\rightarrow$ null $\rightarrow$ while 'done' ch.		

Decrypted text: How old are you?

Example 3:

Text= I am 60 years old.

Shift = right

Shift = right

X=3

Answer:

Encrypted Text: L dp 60 bhduv rog.

Decrypted Text: I am 60 years old.

i	shift	x	string[i]	hangisiye	işlem
0	r	3	I → L	if 4	'I' + 3 = 'L'
1	r	3	-	(x)	(x)
2	r	3	a → d	if 1	'a' + 3 = 'd'
3	r	3	m → p	if 1	'm' + 3 = 'p'
4	r	3	-	(x)	(x)
5	r	3	6	(x)	(x)
6	r	3	0	(x)	(x)
7	r	3	-	(x)	(x)
8	r	3	y → b	if 1 + if 3	'y' + 3 - ('z' - 'a') - 1 = 'b'
9	r	3	e → h	if 1	'e' + 3 = 'h'
10	r	3	a → d	if 1	'a' + 3 = 'd'
11	r	3	r → u	if 1	'r' + 3 = 'u'
12	r	3	s → v	if 1	's' + 3 = 'v'
13	r	3	-	(x)	(x)
14	r	3	o → r	if 1	'o' + 3 = 'r'
15	r	3	l → o	if 1	'l' + 3 = 'o'

15	r	-3		if 1	'd'+3='g'
16	r	3	d → g	(x)	(x)
17	r	3	.		
18	r	3		null → while 'down' w.h.	

Encrypted text: L d p 60 b h d u v r o q.

Decryption ↓

0	r	x-4-1=-3	L → I	if 10	'L'-3='I'
1	r	-3	-	(x)	(x)
2	r	-3	d → a	if 7	'd'-3='a'
3	r	-3	p → m	if 7	'p'-3='m'
4	r	-3	-	(x)	(x)
5	r	-3	b	(x)	(x)
6	r	-3	0	(x)	(x)
7	r	-3	-	(x)	(x)
8	r	-3	b → y	if 7 + if 8	'b'-3 + ('z'-'a')+1
9	r	-3	h → e	if 7	'h'-3='e'
10	r	-3	d → a	if 7	'd'-3='a'
11	r	-3	u → r	if 7	'u'-3='r'
12	r	-3	v → s	if 7	'v'-3='s'
13	r	-3	-	(x)	(x)
14	r	-3	r → o	if 7	'r'-3='o'
15	r	-3	a → i	if 7	'a'-3='i'

15 r -3 0 → d 127 'q' -3 = 'd'
16 r -3 • (x) (x)
17 r -3 '~\0' → while 'don' ch.

Decrypted text: I am 60 years old.

Source Code.

```
#include <stdio.h>
```

```
#define MAX 50
```

```
int main()
```

```
{
```

```
char string[MAX], encrypted_text[MAX];
```

```
char shift, temp;
```

```
int x, i;
```

```
puts("Please enter your sentence.");
```

```
gets(string);
```

```
puts(">>>>Your input sentence is: ");
```

```
puts(string);
```

```
do
{
puts("\nPlease enter shift value. 'l' for left
'r' for right ");
shift=getc(stdin);
}while((shift!='l')&&(shift!='r'));
```

```
do
{
puts("\nPlease enter encryption value.It
has to be above 0.");
scanf("%d",&x);
}while(x<=0);
if(shift=='r')
{
x*=1;
}
else if(shift=='l')
{
x*=-1;
}
```

```
else
{
printf("The shift value is wrong.Exiting...");
exit(-1);
}
puts("[INFO]Encryption is
starting...Loading...");
i=0;
while(string[i]!='\0')
{
if(((string[i]>='a')&&(string[i]<='z')) //if1
{
string[i]+=x;
if((string[i]<'a')) //2
{
string[i]=string[i]+('z'-'a')+1;
}
else if((string[i]>'z')) //3
{
string[i]=string[i]-('z'-'a')-1;
}
```

```
}  
else  
if(((string[i]>='A')&&(string[i]<='Z')) //4  
{  
string[i]+=x;  
if((string[i]<'A')) //5  
{  
string[i]=string[i]+('Z'-'A')+1;  
}  
else if((string[i]>'Z')) //6  
{  
string[i]=string[i]-('Z'-'A')-1;  
}  
}  
i++;  
}  
puts(">>>Encrypted string is: ");  
puts(string);
```

```
puts("[INFO]Decryption is  
starting...Loading...");
```

```
i=0;
x*=-1;
while(string[i]!='\0')
{
if(((string[i]>='a')&&(string[i]<='z')))//7
{
string[i]+=x;
if((string[i]<'a'))//8
{
string[i]=string[i]+('z'-'a')+1;
}
else if((string[i]>'z'))//9
{
string[i]=string[i]-('z'-'a')-1;
}
}
else
if(((string[i]>='A')&&(string[i]<='Z')))//10
{
string[i]+=x;
if((string[i]<'A'))//11
```

```
{  
string[i]=string[i]+('Z'-'A')+1;  
}  
else if((string[i]>'Z')) //12  
{  
string[i]=string[i]-('Z'-'A')-1;  
}  
}  
i++;  
}  
puts(">>>Decrypted string is: ");  
puts(string);  
return 0;  
}
```

